Universität Bonn

Notes for the lecture

Functional Analysis and PDE

held by

Prof. Dr. Stefan Müller

TEXed by

Daniel Emse

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Corrections and improvements

If you have corrections or improvements, contact me via (`daniel.emse@uni-bonn.de`).

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Lecture

Chapter 0

Overview/Teaser

[09.10.2024, Lecture 1]

1. What is FA?
   - Analysis on infinite-dimensional spaces

     examples: $C^0(X), C^\infty(U \subseteq \mathbb{R}^n), \ell_p = \{a : \mathbb{N} \to \mathbb{R} \mid \sum_{n=0}^\infty |a_n|^p < \infty\}$ $(1 \leq p \leq \infty)$
   - Combination of geometric, analytic and (linear) algebraic ideas
2. What is FA good for?
   - Geometry/Topology with manifolds, bundles, loop space
   - PDEs, scalar products
   - Algorithms/Numerics (approxmiation theory; errors)
   - infinite-dimensional representations
   - Mathematical Physics / Quantum Mechanics; spectral theory
   - Probability Theory (infinite products)
3. Why combine PA & PDE?
   - $u : U \subseteq \mathbb{R}^n \to \mathbb{R}$ as a C^2 function:

     $$\Delta u(x) = \sum_{i=1}^n \frac{\partial^2 u}{\partial x_i^2}. \tag{0.1}$$

     with $-\Delta u = f$.

     Classical approach: Given a potential $f \in C^2$, does there exist a solution and is that solution unique?

     Modern approach: $\Delta : X \to Y$ linear map between function spaces. e.g. $X = C^2(U), Y = C^0(U)$. Existance of solutions for all $f \in C^0$ is equiv to surjectivity of $f\Delta$ and uniqueness is equivalent to injectivity.

     Fun fact: C^2 is a bad space for Δ. But Δ is a good operator on many function spaces, e.g. $C^{2,\alpha}$ with $0 < \alpha < 1$, so D^2u is α-Hölder continuous, or for $W^{2,p}$ as Sobolev spaces (where $D^2u \in L^p$ for $1 < p < \infty$), where D^2 is the weak derivative
4. „Big things“
   - Spaces (of founctions) and convergence (norm, scalarproduct)
   - (Bounded) Linear operators, Hahn-Banach theorem, uniform boundedness, ...
   - Hilbert spaces, projections
   - Spectral theory (LA: $A : \mathbb{R}^n \to \mathbb{R}^n$ symmetric, so $\langle Ax, y\rangle = \langle x, Ay\rangle \Rightarrow \exists$ orthogonal basis of eigenvectors; we want something similar in infinite dimensions which makes diagonazing easier, and we can easily compute the inverse or powers of the operator)

Chapter 1

Structures

Generally, all vector spaces will be with $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$.

- on „infinite-dimensional“ spaces
- Convergence and neighbourhoods, topological space
- Metric space (metric, triangle inequality, gives a *distance*)
- Normed space (vector space, $\|\lambda x\| = |\lambda| \cdot \|x\|$ for $\lambda \in \mathbb{R}$, gives a *length*)
- Scalar product space ($\|x\| = \sqrt{\langle x, x \rangle}$, gives a *length and an angle*)

1.1 Topological spaces

Let X a set and 2^X the power set.

Definition 1.1: Topological space, Topology

A *topological space* $(X, \mathcal{T})$ is a pair of a set X and subset $\mathcal{T}$ of 2^X with the following properties:

(i) $\emptyset \in \mathcal{T}, X \in \mathcal{T}$

(ii) finite intersections: $U, V \in \mathcal{T} \Rightarrow U \cap V \in \mathcal{T}$

(iii) arbitrary unions: $U_\alpha \in \mathcal{T}\ \forall \alpha \in A \Rightarrow \bigcup_{\alpha \in A} U_\alpha \in \mathcal{T}$.

The set $\mathcal{T}$ is called a *Topology on* X and the elements in $\mathcal{T}$ are often called *open sets*. A set C is called *closed* if $X \setminus C$ is open.
If $\mathcal{T}_1$ and $\mathcal{T}$ are topologies on X, then $\mathcal{T}_1$ is called *finer* than $\mathcal{T}_2$ or $\mathcal{T}_2$ is called *coarser* than $\mathcal{T}_1$ if $\mathcal{T}_1 \supset \mathcal{T}_2$.

Example 1.2. (i) The finest topology is $\mathcal{T} = 2^X$.

(ii) The coarsest topology is $\mathcal{T} = \{\emptyset, X\}$.

(iii) Standard topology on $\mathbb{R}$: $U \in \mathcal{T}_{st} \iff \forall x \in U \exists \varepsilon > 0 : (x - \varepsilon, x + \varepsilon) \subseteq U$. We will usually use this topology.

(iv) Let $X = \mathbb{R}$ and $\mathcal{T} = \{U \subseteq \mathbb{R} \mid U = \emptyset \text{ or } \mathbb{R} \setminus U \text{ is countable}\}$. Then, the closed sets are countable[1] or $\mathbb{R}$.

(v) Let $(X, \mathcal{T})$ be a topological space. Then, for $A \subseteq X$, the *relative topology* $\mathcal{T}_A$ on A is defined by $\mathcal{T}_A = \{U \cap A \mid U \in \mathcal{T}\}$.

(vi) Let B be a set and $\mathcal{T}_\beta$ a topology on X for all $\beta \in B$. Then, $\mathcal{T} := \bigcap_{\beta \in B} \beta$ defines a topology on X.

[1] We call a set *countable* if it is empty, finite or has an bijection to $\mathbb{N}$.

Given any $\mathcal{S} \subseteq 2^X$,

$$\bigcap_{\substack{\mathcal{T} \text{ top.} \\ \mathcal{T} \supseteq \mathcal{S}}} \mathcal{T} \tag{1.1}$$

is a topology and is called the *coarsest topology containing* $\mathcal{S}$.

(*vii*) *Product topology* of $(X, \mathcal{T}_X), (Y, \mathcal{T}_Y)$ as topological spaces is definied with

$$\mathcal{S} := \{U \times Y \mid U \in \mathcal{T}_X\} \cup \{X \times V \mid V \in \mathcal{T}_Y\}. \tag{1.2}$$

The product topology is $\mathcal{T}_{\mathcal{S}}$, the coarsest topology containing $\mathcal{S}$.

The product topology is the coarsest topology so that the projections $\pi_1(x, y) = x$ and $\pi_2(x, y) = y$ are „continuous“.

Definition 1.3: Interior, Closure, Boundary

Let $(X, \mathcal{T})$ be a topological space and $A \subseteq X$. We define the *interior* A°, the *closure* $\overline{A}$ and the *boundary* ∂A by

$$A^\circ = \bigcup_{\substack{U \subseteq A \\ U \in \mathcal{T}}} U, \quad \overline{A} = \bigcap_{\substack{C \supseteq A \\ X \setminus C \in \mathcal{T}}} C, \quad \partial A = \overline{A} \setminus A^\circ. \tag{1.3}$$

Definition 1.4: Dense subsets, separable topological space

$A \subseteq X$ is *dense* if $\overline{A} = X$. The space X is *separable* if there exists a countable, dense subset.

Definition 1.5: Open neighbourhood

Let $x \in X$. Then, $U \subseteq X$ is an *open neighbourhood of* x if $U \in \mathcal{T}$ and $x \in U$.

Definition 1.6: Continuous maps, Homeomorphisms

Let $(X, \mathcal{T}_X), (Y, \mathcal{T}_Y)$ topological spaces. Then, $f : X \to Y$ is called *continuous* if

$$\forall V \supseteq \{x\} \in \mathcal{T}_Y : \quad f^{-1}(V) = \{x \in X \mid f(x) \in V\} \in \mathcal{T}_X. \tag{1.4}$$

We call f a *homeomorphism* if f is bijective and both f and f^{-1} are continuous.

If $\mathcal{T} = 2^X$, and $f : X \to Y$ is continuous.

If $\mathcal{T} = \{\emptyset, \mathcal{T}\}$, then every constant map is continuous.

[09.10.2024, Lecture 1]
[11.10.2024, Lecture 2]

Definition 1.7: Convergence

Let $(X, \mathcal{T})$ be a topological space and $x : \mathbb{N} \to X$ a sequence. Let $x^* \in X$. We say that x *converges to* x^* if for **every** neighbourhood U of x, the set $\{j \in \mathbb{N} \mid x_j \notin U\}$ is finite.

Example. (*i*) If $\mathcal{T} = \{\emptyset, X\}$, every sequence converges to every point $x^* \in X$.

(*ii*) If $\mathcal{T} = 2^X$, take $U = \{x^*\}$, so only sequences which are constant with up to finitely many exceptions converge.

Definition 1.8: Hausdorff space

A topological space is called a *Hausdorff space* if

(*i*) sets consisting of a single point are closed,

(*ii*) for different points $x \neq y$, there exist disjoint open neighbourhoods: $\exists U_x, U_y \in \mathcal{T}$ with $x \in U_x, y \in U_y$ and $U_x \cap U_y = \emptyset$.

Proposition 1.9. *If $(X, \mathcal{T})$ is Hausdorff, then every sequence has at most one limit point.* □

Definition 1.10: Compactness

A subset K of a topological space $(X, \mathcal{T})$ is *compact* if every open cover of K contains a **finite** subcover: If A is an index set and $U_\alpha \in \mathcal{T}$ for every $\alpha \in A$ and

$$\bigcup_{\alpha \in A} U_\alpha \supseteq K, \tag{1.5}$$

then there exists a **finite** set $A' \subseteq A$ so that

$$\bigcup_{\alpha \in A'} U_\alpha \supseteq K. \tag{1.6}$$

Lemma 1.11: Continuous maps compact to compact

If $f : (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)$ is continuous and $K \subseteq X$ is compact in X, then $f(X)$ is compact in Y.

Proof. Follows directly from the definitions. □

Theorem 1.12: Continuous maps with compact domain attain max and min

If $f : (X, \mathcal{T}_X) \to \mathbb{R}$ is continuous and $K \subseteq X$ is compact, then f attains its minimum and maximum on K, i.e. $\exists \underline{x}, \overline{x} \in K$ so that $f(\underline{x}) = \inf_{x \in K} f(x), f(\overline{x}) = \sup_{x \in K} f(x)$.

Proof. $f(K) \subseteq \mathbb{R}$ is compact, the rest is easy. □

Definition 1.13: Connectedness

Let $(X, \mathcal{T})$ be a topological space. We call X *connected* if it cannot be written as a disjoint union of two open subsets. A subset $A \subseteq X$ is connected if $(A, \mathcal{T}_A)$ is connected.

Exercise 1. Show that intervals in $\mathbb{R}$ are connected.

Theorem 1.14: Intermediate value theorem

If $f : (X, \mathcal{T}_X) \to \mathbb{R}$ is continuous and $(X, \mathcal{T}_X)$ is connected, and there exist $a, b \in X$ so that $f(a) < 0, f(b) > 0$. Then, there exists an $x \in X$ with $f(x) = 0$.

Proof. Assume $f(x) \neq 0$ for all $x \in X$. We know that $U := f^{-1}((0, \infty))$ is open, and $X \setminus U = f^{-1}((-\infty, 0]) = f^{-1}((-\infty, 0))$ is also open. Then, $X = U \cup (X \setminus U) \Rightarrow X$ not connected. ↯ □

1.2 Metric spaces

1.2.1 Definition and examples

Let X be a set and $d : X \times X \to [0, \infty)$.

Definition 1.15: Metric space, metric

(X, d) is called a *metric space* and d is called *metric* if

(*i*) $d(x, y) = 0 \Leftrightarrow x = y$,

(*ii*) $\forall x, y : d(x, y) = d(y, x)$,

(*iii*) (triangle inequality) $d(x, y) \leq d(x, z) + d(z, y)$ for all $x, y, z \in X$.[1]

If only (*ii*) and (*iii*) are fulfilled, we call d a *semimetric* or *pseudometric*, and with $x \sim y :\Leftrightarrow d(x, y) = 0$, we get a metric $\overline{d}$ on the equivalence classes.

[1] „Where the triangle inequality is not true is on airline flights. Sometimes, it is cheaper to fly from Köln/Bonn to Mallorca first before flying to mainland Spain later.“ – Stefan Müller, 11.10.2024

Lemma 1.16: Composition of concave map and metric is metric

Assume $\psi : [0, \infty) \to [0, \infty)$ is continuous, non-decreasing, ***concave*** *and $\psi(0) = 0, \psi(x) > 0$ for $x > 0$. If d is a metric on X, so is $\psi \circ d$.*

Proof. It is enough and easy to check that $\psi(a + b) \leq \psi(a) + \psi(b)$. □

Example. Define $\psi(t) := \frac{t}{1+t} = 1 - \frac{1}{1+t}$. Then, $\psi \in C^0([0, \infty)) \cap C^2((0, \infty))$. Non-decreasing and concave are equivalent to $\psi' \geq 0$ and $\psi'' \leq 0$.

Example. (*i*) Let $X = \mathbb{R}^n$ and $d_\infty(x, y) = \max_{1 \leq i \leq n} |x_i - y_i|$.

(*ii*) $X = \mathbb{R}^n, d_p(x, y) := (\sum_{i=1}^n |x_i - y_i|^p)^{\frac{1}{p}}$ for $1 \leq p < \infty$,

(*iii*) $\ell_p = \left\{ a : \mathbb{N} \to \mathbb{R} \;\middle|\; (\sum_{i=1}^\infty |x_i|^p)^{\frac{1}{p}} < \infty \right\}$ and $d_p(a, b) := (\sum_{i=1}^\infty |a_i - b_i|^p)^{\frac{1}{p}}$ for $1 \leq p < \infty$

(*iv*) $\ell_\infty := \{a : \mathbb{N} \to \mathbb{R} \mid \sup_{i \in \mathbb{N}} |a_i| < \infty\}$ with $d_\infty(a, b) := \sup_{i \in \mathbb{N}} |a_i - b_i|$

(*v*) $X = \mathbb{R}$ with $d(x, y) = \frac{|x-y|}{1+|x-y|} = \psi(|x - y|)$ for $\psi(t) = \frac{t}{1+t}$. By Lemma 1.16, d is a metric on $\mathbb{R}$. Fun fact: $0 \leq d(x, y) < 1$.

(*vi*) Metric on the space of all sequences:

$$d(a, b) = \sum_{i=0}^\infty 2^{-i} \frac{|a_i - b_i|}{1 + |a_i - b_i|} \leq \sum_{i=0}^\infty 2^{-i} \leq 2. \tag{1.7}$$

(*vii*) Let X be a set and (Y, d_Y) a metric space and $f : X \to Y$ **injective**. Then, we can define the *pullback metric* $d_X(x, y) := d_Y(f(x), f(y))$. d_X is a metric on X.

(*viii*) $\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, \infty\}$. Then,

$$f : \overline{\mathbb{R}} \to [-1, 1], x \mapsto \begin{cases} -1 & x = -\infty \\ \frac{x}{1+|x|} & x \in \mathbb{R} \\ 1 & x = \infty \end{cases} \tag{1.8}$$

is bijective. Then, $d_{\overline{\mathbb{R}}}(x, y) = |f(x) - f(y)|$ is a pullback metric.

(*ix*) Shortest path on a graph

Remark. Let (X,d) be a metric space, let $A,B \subseteq X$, let $x \in X$ and let $r > 0$. Further distances:

Diameter of a subset

$$\sup\{d(x,y) \mid x,y \in A\} \tag{1.9}$$

(with the convention $\operatorname{diam}(A) = \infty$ if $\{d(x,y) \mid x,y \in A\}$ is unbounded)

Distance between two sets

$$\operatorname{dist}(A,B) = \inf\{d(x,y) \mid x \in A, y \in B\} \tag{1.10}$$

Distance between a point and a set

$$\operatorname{dist}(x,A) = \inf\{d(x,y) \mid y \in B\} = \operatorname{dist}(\{x\},A) \tag{1.11}$$

r-neighbourhood of a set

$$B_r(A) := \{x \in X \mid \operatorname{dist}(x,A) < r\} \tag{1.12}$$

r-ball of a point

$$B_r(x) = B(x,r) = \{y \in X \mid d(y,x) < r\} \tag{1.13}$$

Hausdorff distance of two sets

$$d_H(A,B) := \inf\{r > 0 \mid A \subseteq B_r(B) \text{ and } B \subseteq B_r(A)\} \tag{1.14}$$

and $d_H(\{x\},\{y\}) = d(x,y)$.

Remark. It is easy to show that

$$d_H(A,B) = \max\left(\sup_{x\in A}\operatorname{dist}(x,B), \sup_{y\in B}\operatorname{dist}(y,A)\right). \tag{1.15}$$

Proof. Exercise. □

Proposition 1.17 (Topology induced by a metric)**.** *Let (X,d) be a metric space. Let $\mathcal{T}_d$ consist of all sets U with the following property:*

$$\forall x \in U \exists \varepsilon > 0 : B_\varepsilon(x) \subseteq U. \tag{1.16}$$

Then, $\mathcal{T}_d$ is a topology on X.

Proof. Clear. For the intersection, take the ball with the smaller radius. □

Lemma 1.18: Hausdorff distance is metric on closed and bounded sets

Let (X,d) be a Hausdorff space. Then, d_H defines a metric on

$$\mathcal{C} := \{A \subseteq X \mid \emptyset \neq A \text{ is closed and bounded}\}. \tag{1.17}$$

Proof. Exercise. □

[11.10.2024, Lecture 2]
[16.10.2024, Lecture 3]

1.2.2 Convergence and continuity in metric spaces

Proposition 1.19. *Let (X,d) be a metric space and $x:\mathbb{N}\to X$ a sequence, and $x^*\in X$. Then, the following two statements are equivalent:*

(i) *x converges to x^* in $(X,\mathcal{T}_d)$,*

(ii) $\lim_{n\to\infty} d(x_n,x^*)=0$.

Moral. „In metric spaces, anything can be done by sequences“

Proposition 1.20. *Let (X,d) be a metric space and let $A\subseteq X$. Then, the following two statements are equivalent:*

(i) *A is closed, so $X\setminus A\in\mathcal{T}_d$,*

(ii) *A is* sequentially closed*, so for every sequence $x:\mathbb{N}\to A$ which converges to $x^*\in X$, one has $x^*\in A$.*

Moreover, the closure of any set A is given by the sequential closure*, this means*

$$\overline{A}=\{x^*\in X\mid \exists x:\mathbb{N}\to A, x \text{ converges to } x^*\}. \tag{1.18}$$

Proof. Easy exercise. □

Proposition 1.21. *Let $(X,d_X),(Y,d_Y)$ be two metric spaces and $f:X\to Y$. Then, the following three statements are equivalent:*

(i) *f is continuous in as a map $f:(X,\mathcal{T}_{d_X})\to(Y,\mathcal{T}_{d_Y})$,*

(ii) *For every $x\in X,\varepsilon>0$ there exists a $\delta>0$ so that $d_X(z,x)<\delta\Rightarrow d_Y(f(z),f(x))<\varepsilon$,*

(iii) *For every sequence x convergent to x^*, also $f(x)$ converges to $f(x^*)$.*

We know that every metric space is Hausdorff. *Question:* If $(X,\mathcal{T})$ is a Hausdorff space, does there exist a metric $d:X\times X\to[0,\infty)$ such that $\mathcal{T}=\mathcal{T}_d$? Or shorter, is X metrizable?

If X is second countable (so there exist countable neighbourhoods), X is metrizable, and then, X ist first countable.

Definition 1.22: Equivalent metrices, stronger metrices

Let d_1 and d_2 be metrices on X.
We say d_1 *is stronger than* d_2 if $\mathcal{T}_{d_1}\supset\mathcal{T}_{d_2}$.
We say d_1 *is equivalent to* d_2 if $\mathcal{T}_{d_1}=\mathcal{T}_{d_2}$.

Proposition 1.23. *The following statements are equivalent:*

(i) *d_1 is stronger than d_2,*

(ii) *The identity map $x\mapsto x$ for $(X,d_1)\to(X,d_2)$ is continuous,*

(iii) *Every sequence that converges in d_1 also converges in d_2,*

(iv) *$\forall x\in X\forall\varepsilon>0\exists\delta>0$ so that $d_1(x,y)<\delta\Rightarrow d_2(x,y)<\varepsilon$.*

Example.

(i) Let $X=\mathbb{R}$ and $d_1(x,y)=|x-y|, d_2=|\arctan(x)-\arctan(y)|$. Then, d_1 and d_2 are equivalent (Exercise 1.4).

(*ii*) Let $\ell_1 := \{x : \mathbb{N} \to \mathbb{R} \mid \sum_{k=0}^{\infty} |x_k| < \infty\}$ with $d_1(x,y) := \sum_{k=0}^{\infty} |x_k - y_k|$ and $d_\infty(x,y) = \sup_{k\in\mathbb{N}} |x_k - y_k| \le d_1(x,y)$. We see that d_1 is stronger than d_∞, but not equivalent: See the map

$$\mathbb{N} \to (\mathbb{N} \to \mathbb{R}), \quad j \mapsto x_k^{(j)} := \begin{cases} 0 & k < j-1 \text{ or } k > 2j \\ \frac{1}{k} & j \le k \le 2j. \end{cases} \tag{1.19}$$

Then, $\left|x_k^{(j)}\right| \le \frac{1}{j}$ with $d_\infty(x^{(j)}, 0) \le \frac{1}{j} \to 0$ as $j \to \infty$, but $d_1(x_k^{(j)}, 0) = \sum_{k=0}^{\infty} x_k^{(j)} = \sum_{k=j}^{2j} \frac{1}{k} \ge j\frac{1}{2j} = \frac{1}{2}$.

Definition 1.24: Separable metric space

A metric space (X,d) is called *separable* if $(X, \mathcal{T}_d)$ is separable.

1.2.3 Completeness

Definition 1.25: Cauchy sequence, completeness

Let (X,d) be a metric space.

(*i*) A sequence $x : \mathbb{N} \to X$ is a *Cauchy sequence* if $\forall \varepsilon > 0 \exists k_0$ so that for all $k, \ell \ge k_0$: $d(x_k, x_\ell) < \varepsilon$,

(*ii*) (X,d) is called *complete* if every Cauchy sequence converges.

Example.

(*i*) $\mathbb{Q}$ equipped with the standard metric is not complete.

(*ii*) (ℓ_p, d_p) are complete for all $1 \le p \le \infty$.

(*iii*) $\mathbb{R}$ equipped with the standard metric is complete, but not with the metric $\tilde{d}(x,y) := |\arctan(x) - \arctan(y)|$, although they induce the same topology (Exercise 1.4).

(*iv*) $\ell_1 = \{x : \mathbb{N} \to \mathbb{R} \mid \sum_{k=0}^{\infty} |x_k| < \infty\}$ with d_1 is complete (see (*ii*)), but (ℓ_1, d_∞) is not complete:

Take $j \mapsto x_k^{(j)}$ and set $x_k^{(j)} := \begin{cases} \frac{1}{k} & 1 \le j \le k \\ 0 & \text{else} \end{cases}$. For any $m > j$, we get $\left|x_k^{(j)} - x_k^{(m)}\right| \le \frac{1}{j}$, so $d_\infty(x^{(j)}, x^{(m)}) \le \frac{1}{j}$ for $m > j$, so the sequence is Cauchy but has no limit.

Proposition 1.26. *Let (X,d) be a complete metric space and $A \subseteq X$ closed. Let $d_A := d|_{A\times A}$. Then, (A, d_A) is complete.*

Proof. If $x : \mathbb{N} \to A$ is a Cauchy sequence in X, by completeness, $x_k \to x^*$ in X. As A is (sequentially) closed, $x^* \in A$. □

Definition 1.27: Isometric immersion, isometry

Let (X, d_X) and (Y, d_Y) be metric spaces. A map $f : X \to Y$ is called an *isometric immersion* if

$$d_Y(f(x), f(y)) = d_X(x,y) \tag{1.20}$$

for all $x, y \in X$. The map f is called an *isometry* if it is in addition bijective

Remark. An isometric immersion is automatically bijective. Thus, if f is an isometric immersion, than f is an isometry from X to $f(X)$. The spaces X and $f(X)$ (equipped with the restriction of the metric d_Y) are indistinguishable as metric spaces. In particular, a sequence $f : \mathbb{N} \to \mathbb{R}$ is convergent in X if and only if $f \circ x$ is convergent in $f(X)$, and X is a Cauchy sequence in X if and only if $f \circ x$ is a Cauchy sequence in $f(X)$.

Theorem 1.28: Completion

Let (X, d) be a metric space. Then, there exists a complete, metric space $(\tilde{X}, \tilde{d})$ and an isometric immersion $j : X \to \tilde{X}$ such that $j(X)$ is dense in $\tilde{X}$.

The space $\tilde{X}$ is called the *completion of* X (and is unique up to isometries).

Proof. We will see a short proof later. Here we just sketch the standard proof which is based on considering the space of Cauchy sequences in X modulo converging sequences. This proof is modelled on the construction of the real numbers $\mathbb{R}$ from the rational numbers $\mathbb{Q}$ (see e.g. Alt's book for details).

Let

$$\hat{X} := \{x : \mathbb{N} \to X \mid x \text{ Cauchy sequence}\}. \tag{1.21}$$

Define

$$x \sim y \quad :\Longleftrightarrow \quad \lim_{k\to\infty} d(x_k, y_k) = 0. \tag{1.22}$$

It is easy to see that $\sim$ is an equivalence relation. The equivalence class of $x \in \hat{X}$ is defined by

$$[x] := \left\{ z \in \hat{X} \;\middle|\; z \sim x \right\}, \tag{1.23}$$

and the space of equivalences is defined by $\hat{X} := {}^{X}/_{\sim}$. On $\hat{X}$, we define

$$\tilde{d}([x], [y]) := \lim_{j\to\infty} d(x_j, y_j). \tag{1.24}$$

It follows from the definition of the equivalence classes and the triangle inequality that $\tilde{d}$ is well-defined, i.e., the right hand side only depends on the equivalence classes of the Cauchy sequences x and y. Since x and y are Cauchy sequences, it is also easy to see that $j \mapsto d(x_j, y_j)$ is a Cauchy sequence in $\mathbb{R}$ and that therefore $\lim_{j\to\infty} d(x_j, y_j)$ exists. It is easy to see that $\tilde{d}$ is a metric on $\tilde{X}$. Then, one shows that

- $(\tilde{X}, \tilde{d})$ is complete,
- the map $j : X \to \tilde{X}$ which maps a to the (equivalence class of the) constant sequence with value a is an isometry
- $j(X)$ is dense in $(\tilde{X}, \tilde{d})$

□

Lemma 1.29: Hausdorff metric space is complete

Let (X, d) be a complete, metric space. Then, the space $(\mathcal{C}, d_H)$ of closed, bounded, non-empty set with the Hausdorff metric (introduced in Lemma 1.18) is complete.

Proof. Exercise. *Hint:* First show that a decreasing Cauchy sequence of sets B_j converges to $B^* := \bigcap_{j\in\mathbb{N}} B_j$ (and in particular, $B^* \neq \emptyset$). By passage to a subsequence, you can assume without loss of generality that $d_H(B_j, B_{j+1}) < 2^{-j}$ (explain why). For a general Cauchy sequence A_j define $B_j := \overline{\bigcup_{k\geq j} A_j}$ and show that $d_H(A_j, B_j) \to 0$. □

One can also introduce a metric on set of (equivalence classes of) bounded and complete metric spaces, the so called *Gromov-Hausdorff metric*. Here we say that $(X, d_X) \sim (Y, d_Y)$ if there exist an isometry $I : X \to Y$. Then, one defines

$$\begin{aligned} &d_{GH}((X, d_X), (Y, d_Y)) \\ &:= \inf \{d_{H,Z}(I(X), J(Y)) \mid (Z, d_Z) \text{ metric space, } I : X \to Z, J : Y \to Z \text{ isom. immers.}\} \end{aligned} \tag{1.25}$$

where $d_{H,Z}$ denotes the Hausdorff distance in the metric space (Z, d_Z). The GH metric plays an important role in geometry as well as in application to image processing.

[16.10.2024, Lecture 3]
[18.10.2024, Lecture 4]

1.3 Normed spaces

Let X be a $\mathbb{K}$-vector space. Recall our convention that $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$.

Definition 1.30: Norm

A pair $(X, \|\cdot\|)$ is called *normed space* if X is a $\mathbb{K}$-vector space and $\|\cdot\| : X \to [0, \infty)$ with the following properties:

(*i*) Definiteness: $\|x\| = 0 \Rightarrow x = 0$

(*ii*) Homogeneity: $\|\lambda x\| = |\lambda| \cdot \|x\|$ for all $\lambda \in \mathbb{K}, x \in X$

(*iii*) Triangle inequality: $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in X$.

If only (*ii*) and (*iii*) hold, we call $\|\cdot\|$ a *seminorm*.

If $(i) - (iii)$ hold, the map d defined by $d(x, y) = \|x - y\|$ is a metric that is translation invariant.

It follows from the triangle inequality that

$$|\|x\| - \|y\|| \leq \|x - y\| \tag{1.26}$$

That implies, that the map $X \to \mathbb{R}, x \mapsto \|x\|$ is continuous (actually, Lipschits continuous with Lipschitz constant 1).

Definition 1.31: Banach space

A normed space which is complete is called *Banach space*.

Example. For $1 \leq p \leq \infty$, the sequence spaces ℓ_p are Banach spaces if equipped with norms $\|x\|_p := (\sum_{k=0}^{\infty} |x_k|^p)^{\frac{1}{p}}$ for $p < \infty$ and $\|x\|_\infty := \sup_{k \in \mathbb{N}} |x_k|$ respectively.

Definition 1.32: Stronger norms, equivalent norms

Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be norms on a $\mathbb{K}$-vector space X. We say that $\|\cdot\|_1$ is *stronger than* $\|\cdot\|_2$ if the corresponding metric d_1 is stronger than d_2.
We say that $\|\cdot\|_1$ and $\|\cdot\|_2$ *are equivalent* if d_1 and d_2 are equivalent.

Proposition 1.33. *Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be norms on a $\mathbb{K}$-vector space X. Then, $\|\cdot\|_1$ is stronger that $\|\cdot\|_2$ iff there exists a $C > 0$ such that*

$$\forall x \in X : \quad \|x\|_2 \leq C \cdot \|x\|_1 . \tag{1.27}$$

Moreover, $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent iff three exist $c, C > 0$ such that

$$\forall x \in X: \quad c\|x\|_1 \leq \|x\|_2 \leq C\|x\|_1. \tag{1.28}$$

Proof. The second statement follows immediately from the first statement.

Recall that d_1 is stronger than d_2 iff the identity map from (X, d_1) to (X, d_2) is continuous. If (1.27) holds, the identity map is continuous.

Now assume that $\mathrm{id}: (X, d_1) \to (X, d_2)$ is continuous. Apply the ε-δ definition of continuity at the point $x = 0$:

$$\exists \delta > 0: \quad \forall y: \quad \|y\|_1 < \delta \Rightarrow \|y\|_2 < 1. \tag{1.29}$$

Let $C > 0$ be any number such that $C > \frac{1}{\delta}$. Let $X \in X \setminus \{0\}$. Define $y := \frac{x}{C\|x\|_1} \Rightarrow \|y\|_1 = \frac{1}{C} < \delta$. This implies that $\|y\|_2 < 1$. Now, $x = C\|x\|_1 y$ and thus $\|x\|_2) C\|x\|_1 \|y\|_2 \leq C\|x\|_1$. □

Example. (*i*) On $\mathbb{R}^n$, all norms are equivalent. (Analysis II)

(*ii*) See $\ell_1 := \{x: \mathbb{N} \to \mathbb{R} \mid \sum_{k=1}^{\infty} |x_k| < \infty\}$ with $\|x\|_1 = \sum_{k=1}^{\infty}, |x_k|$ and $\|x\|_\infty = \sup_{k\in\mathbb{N}} |x_k|$. We have $\|x\|_\infty \leq \|x\|$, so $\|\cdot\|_1$ is stronger than $\|\cdot\|_\infty$ but $\|\cdot\|_\infty$ is not equivalent to $\|\cdot\|_1$.

1.4 Hilbert spaces

For $\mathbb{C} \ni \alpha = a + bi$ with $a, b \in \mathbb{R}$, we say $\overline{\alpha} := a - bi$ is the *complex conjugate*. We get $\alpha\overline{\alpha} = a^2 + b^2$.

Definition 1.34: Sesquilinear form

Let X be a $\mathbb{K}$-vector space. A map $X \times X \to \mathbb{K}, (x_1, x_2) \mapsto \langle x_1, x_2\rangle_X$ is called a *sesquilinear form* if for all $x, y, x_1, x_2, y_1, y_2 \in X$ and for alle $\alpha \in \mathbb{K}$, we have

(*i*) $\langle \alpha x, y\rangle_X = \alpha \langle x, y\rangle_X$ and $\langle x, \alpha y\rangle_X = \overline{\alpha} \langle x, y\rangle_X$,

(*ii*) $\langle x_1 + x_2, y\rangle_X = \langle x_1, y\rangle_X + \langle x_2, y\rangle_X$, $\langle x, y_1 + y_2\rangle_X = \langle x, y_1\rangle_X + \langle x, y_2\rangle_X$.

A sesquilinear form is *symmetric* if

$$\langle y, x\rangle_X = \langle x, y\rangle_X \qquad \forall x, y \in X. \tag{1.30}$$

A sesquilinear form is *positive semidefinite* if

$$\langle x, x\rangle_X \geq 0 \qquad \forall x \in X, \tag{1.31}$$

and *positive definite* if $\langle x, x\rangle_X \geq 0$ and $\langle x, x\rangle_= 0 \Rightarrow X = 0$ for all $x \in X$.

Example. (*i*) $X = \mathbb{C}$ and $\langle x, y\rangle = x\overline{y}$,

(*ii*) $X = \mathbb{C}^n$ and $\langle x, y\rangle = \sum_{i=1}^n x_i \overline{y_i}$,

(*iii*) $X = (\ell_2, \mathbb{C})$ and $\langle x, y\rangle = \sum_{i\in\mathbb{N}} x_i \overline{y_i}$.

Definition 1.35: Scalar product

A positive definite, symmetric sesquilinear form on a $\mathbb{K}$-vector space is a *Scalar product.*. Then, the space $(X, \langle\cdot,\cdot\rangle_X)$ is called a *Pre-Hilbert space*.

What are the key features of a scalar product?

- Geometrically: Orthogonal if $\langle x, y\rangle_X = 0$; orthogonal projections; Pythagoras

- Analytically: Cauchy-Schwarz inequality

Lemma 1.36: Scalar product traits

Let (X) be a $\mathbb{K}$-vector space and $\langle\cdot,\cdot\rangle_X$ a positive semidefinite, symmetric sesquilinear form and define

$$\|x\| := \sqrt{\langle x,x\rangle_X}. \tag{1.32}$$

Then,

(i) Homogeneity: $\|\alpha x\| = |\alpha|\cdot\|x\|$,

(ii) Cauchy-Schwarz inequality: $|\langle x,y\rangle| \le \|x\|\cdot\|y\|$,

(iii) Triangle inequality: $\|x+y\| \le \|x\| + \|y\|$,

(iv) Parallelogram identity: $\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|x\|^2)$.

In particular, $\|\cdot\|$ is a seminorm. If, in addition, $\langle\cdot,\cdot\rangle_X$ is positive definite, then $\|\cdot\|$ is a norm.

Proof. (i) clear by definition.

(ii) Without loss of generality, assume $\langle x,y\rangle \neq 0$.

Special case: $\langle x,y\rangle \in \mathbb{R}_+$. Assume in addition that $\|x\| \neq 0, \|y\| \neq 0$. Set $\xi := \frac{x}{\|x\|}$ and $\eta := \frac{y}{\|y\|}$, so $\|\xi\| = \|\eta\| = 1$. We have

$$\begin{aligned} &0 \le \langle \xi-\eta, \xi-\eta\rangle = \|\xi\|^2 + \|\eta\|^2 - \langle\xi,\eta\rangle - \langle\eta,\xi\rangle = \|\xi\|^2 + \|\eta\|^2 - 2\mathrm{Re}\,\langle\xi,\eta\rangle \\ \iff\ &1 = \frac{1}{2}\left(\|\xi\|^2 + \|\eta\|^2\right) \ge \mathrm{Re}\,\langle\xi,\eta\rangle = \frac{\mathrm{Re}\,\langle x,y\rangle}{\|x\|\,\|y\|}\cdot|\langle x,y\rangle| = \mathrm{Re}\,\langle x,y\rangle \le \|x\|\,\|y\|\,. \end{aligned} \tag{1.33}$$

If $\|x\| = 0$, apply the calculation with $\xi = kx$ and $\eta = \frac{1}{k}y$. Then, $|\langle x,y\rangle| = |\langle\xi,\eta\rangle| \le \frac{\|y\|^2}{k^2}$ for all $k > 0$ which implies $\langle x,y\rangle = 0$. A similar argument applies if $\|y\| = 0$.

From special case to general case: Assume $\langle x,y\rangle \neq 0$. Define $\alpha := \frac{\langle x,y\rangle}{|\langle x,y\rangle|}$, so $\|\alpha\| = 1$, and define $\tilde{x} := \overline{\alpha}\cdot x$, so $\langle\tilde{x},y\rangle = \overline{\alpha}\cdot\langle x,y\rangle = \overline{\alpha}\alpha\,|\langle x,y\rangle| = |\langle x,y\rangle| > 0$.

By the special case,

$$|\langle x,y\rangle| = |\langle\tilde{x},y\rangle| \le \|\tilde{x}\|\cdot\|y\| = \|x\|\cdot\|y\|\,. \tag{1.34}$$

(iii) Exercise.

(iv) Exercise.

□

Lemma 1.37: Scalar product from norm

Let $(X,\|\cdot\|)$ be a normed space. Then, there exists a scalar product $\langle\cdot,\cdot\rangle$ on X such that $\|x\| = \sqrt{\langle x,x\rangle}$ iff $\|x+y\|^2 + \|x-y\|^2 = 2\left(\|x^2\| + \|y\|^2\right)$.

Proof. Exercise. The *only if* part is Lemma 1.36.(iv). □

Definition 1.38: Hilbert space

A pre-Hilbert space which is complete in the norm induced by the scalar product is called a *Hilbert space*.

Chapter 2

Function spaces

2.1 Spaces of bounded, continuous and differentiable functions

Definition 2.1: $B(X;Y)$: Space of bounded functions

Let X be a set and $(Y, \|\cdot\|_Y)$ be a normed space. The *space of bounded functions* $B(X;Y)$ is definied by

$$B(X;Y) := \left\{ f : X \to Y \;\middle|\; \sup_{x\in X} \|f(x)\|_Y < \infty \right\}. \tag{2.1}$$

We define

$$\|f\|_{B(X;Y)} := \sup_{k\in\mathbb{N}} \|f(x)\|_Y . \tag{2.2}$$

Proposition 2.2. *The Space $(B(X;Y), \|\cdot\|_{B(X;Y)})$ is a normed space. If Y is a Banach space, then, $B(X;Y)$ is a Banach space.*

Proof. Exercise. □

If $Y = \mathbb{R}$ (or $\mathbb{C}$), we write $B(X) := B(X, \mathbb{R})$.

Example. (*i*) If $X = \{1, \dots, n\}$, we have $B(X;\mathbb{R}) = \mathbb{R}^n$ and $B(X;Y) = \underbrace{Y \times \cdots \times Y}_{n \text{ times}}$.

(*ii*) If $X = \mathbb{N}$ and $Y = \mathbb{R}$, we have $B(\mathbb{N};\mathbb{R}) = \ell_\infty$, the space of bounded sequences.

[18.10.2024, Lecture 4]
[23.10.2024, Lecture 5]

[23.10.2024, Lecture 5]
[25.10.2024, Lecture 6]

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